

Clickbait and Emotional Framing: An Analysis of YouTube News Channels

Introduction

YouTube has become a major platform for news consumption, where algorithmic visibility rewards emotionally charged and attention-grabbing content. Clickbait and emotional framing have emerged as central strategies for maximizing engagement, often relying on fear, anger, and sensational language (Blom & Hansen, 2014). These techniques raise concerns about polarization and the transformation of news discourse into affect-driven communication.

Mainstream news outlets such as BBC News and CNN generally adhere to institutional journalistic norms, while alternative channels like The Young Turks and The Jimmy Dore Show adopt more opinionated and polemical styles. Prior research suggests that alternative media rely heavily on emotional framing to foster ideological alignment and viral sharing (Boberg et al., 2020). YouTube algorithms favor high-engagement content, reinforcing polarization. Clickbait has been linked to increased shares through high-emotional headlines (Mukherjee et al., 2022).

This study addresses the following research question: *How do alternative and mainstream news channels differ in their use of clickbait and emotional language on YouTube, and how does this relate to audience engagement and interaction?*

We hypothesize that alternative channels use stronger negative emotional framing (fear and anger), generating higher but more volatile engagement, while mainstream channels display more stable reactions through neutral framing.

Methodology

A mixed-methods approach combining text and network analysis was used. All analyses were conducted in Python using Pandas, Gensim (LDA), VADER (sentiment), NRCLEX (emotion detection), and NetworkX (network metrics).

Data Collection

Data were collected via the YouTube Data API v3 from four channels: BBC News and CNN (mainstream), and The Young Turks and The Jimmy Dore Show (alternative). The main dataset spans four months (October 2025–January 2026) and includes 2,608 videos and 451,047 comments. Video metadata included title, description, publication date, and engagement metrics. Comment data included text, author, likes, and timestamps.

Due to API limitations, network analysis focused on the most recent 15 days (January 5–20, 2026). For each channel, replies were retrieved for the top 1,500 comments ranked by likes (maximum 20 replies each), yielding 10,456 replies.

Preprocessing and Operationalization

Text was cleaned using standard NLP preprocessing (SpaCy).

Clickbait was operationalized as the use of sensationalist and curiosity-driven language in video titles aimed at maximizing attention through emotional or hyperbolic cues (Blom & Hansen, 2014). It was identified using a combination of lexical and structural indicators, including forward-referencing expressions (e.g., “this changes everything”), superlatives and intensifiers (e.g., “shocking”, “epic”) (Pengnate et al., 2021), emotional trigger words (e.g., “exposed”, “attack”, “lie”), and list-based formulations. Structural features such as punctuation intensity (exclamation and question marks), uppercase character ratio, and title length were also included as proxies of clickbait style (Chakraborty et al., 2016).

Emotional language was measured through two complementary approaches: sentiment polarity (VADER) and discrete emotion detection using NRCLEX, which assigns intensity scores to eight basic emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust).

Analytical Techniques

The analysis consisted of three phases:

1. *Exploratory Data Analysis (EDA)*: temporal patterns of video production and comment activity; engagement distributions via boxplots; computation of clickbait indicators (sensational terms, uppercase ratio, title length).
2. *Linguistic Analysis*: word frequencies, topic modeling (LDA with five topics per channel), sentiment and emotion dominance.
3. *Network Analysis*: directed reply networks with nodes as users and edges as replies; metrics included degree, density, and connected components. Visualization was restricted to nodes with degree ≥ 5 (max 200 nodes per channel).

Results and Discussion

This study examined whether emotional and clickbait-oriented framing in video titles and descriptions translates into distinct patterns of audience engagement and emotional reaction across mainstream (BBC News, CNN) and alternative channels (The Young Turks, The Jimmy Dore Show). The results reveal systematic differences in framing strategies and in the structure and emotional alignment of user interactions.

Analyzing engagement patterns reveals a nuanced landscape that defies a simple mainstream versus alternative dichotomy. In terms of total volume, engagement is polarized around the market leaders of each category: *CNN* (Mainstream) and *The Young Turks* (Alternative) accumulate the highest number of total comments. However, when examining

average engagement per video, alternative channels consistently show higher values, indicating a more mobilized and reactive core audience.

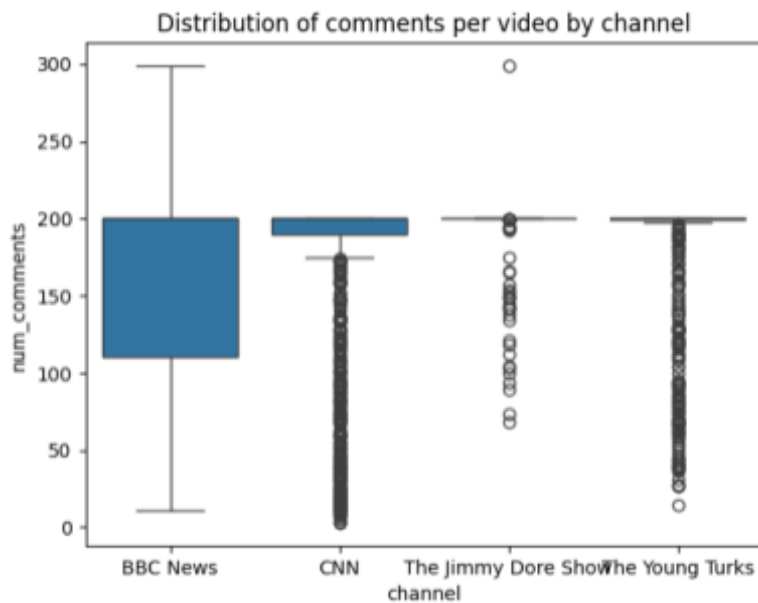


Figure 1: Distribution of engagement per video across channels.

Boxplot analysis (*Fig. 1*) shows that BBC News exhibits the highest variability in comment counts across videos, with a wide interquartile range and long whiskers, indicating organic fluctuations.

In contrast, CNN, The Young Turks, and The Jimmy Dore Show show much narrower boxes around a relatively high baseline among typical videos but pronounced left-skewed distributions with many low outliers. This pattern is characteristic of a hit driven model: most videos receive a fairly similar number of

comments, while a smaller subset flops badly, generating lower or even zero engagement.

Clickbait indicators confirm sharper stylistic contrasts: alternative channels employ more lexical and structural clickbait features (exclamations, uppercase usage, emotionally charged terms). However, these features show only weak correlations with comment volume per video, suggesting that engagement is not driven by formal clickbait alone but by thematic and emotional framing, particularly around polarizing political topics (e.g., Trump, Israel, war, scandal).

Linguistic and topic modeling analyses further show that mainstream channels adopt an institutional and event-driven framing (war, geopolitics, scandals) with relatively neutral lexical style, whereas alternative channels emphasize polemical and personalized narratives (“expose,” “lie,” “attack,” “assassination”) and host-centered discourse. Titles in alternative channels are therefore more explicitly designed to evoke indignation and moral outrage, while descriptions shift toward branding, community building, and monetization.

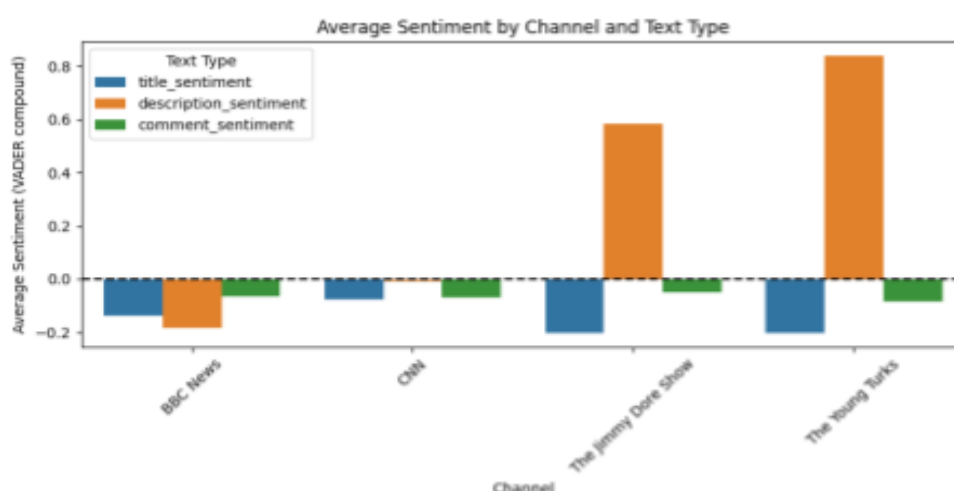


Figure 2: Emotional distributions (continuous score) in titles, descriptions, comments by channel.

Sentiment and emotion analysis reveal that fear and anger dominate titles across all channels, confirming the centrality of emotional framing in news production. A critical divergence emerges in descriptions (*Fig. 2*): mainstream channels maintain coherence between titles and descriptions, whereas alternative channels display a pronounced emotional split. Their titles stress fear, anger, and surprise, while descriptions are dominated by trust, anticipation, and joy, reflecting strategies of branding and audience retention.

User comments exhibit higher emotional intensity than titles regardless of channel, with trust paradoxically emerging as the dominant emotion alongside fear and anger. The bimodal distribution of comment sentiment scores, with peaks near neutral and moderate negative values, further reinforces the polarized nature of audience reactions. This indicates that emotional framing does not simply provoke hostility but also strengthens perceived legitimacy and group belonging, particularly in alternative channels where indignation in titles is transformed into supportive discourse in comments.

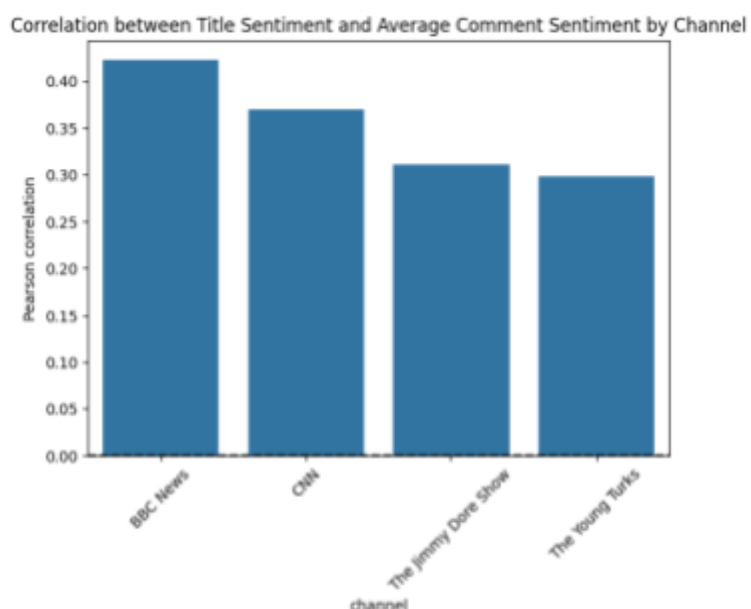


Figure 3: Correlation between title sentiment and average comment sentiment

The alignment between production and reaction differs by channel type. Correlation analyses (*Fig. 3*) show stronger coherence between title sentiment and average comment sentiment in mainstream channels ($r \approx 0.37\text{--}0.42$) than in alternative channels ($r \approx 0.30$). This suggests that alternative audiences respond less to individual titles and more to pre-existing ideological identities, consistent with echo-chamber dynamics.

Emotion–engagement correlations further indicate that

no single emotion robustly predicts comment volume. Fear and anger show weak or inconsistent relationships with engagement, and in alternative channels correlations are often negative. Emotional intensity thus amplifies polarization rather than uniformly increasing participation (Bakir & McStay, 2017).

Network analysis highlights structural differences in interaction patterns (*Fig. 4*). BBC News generates the largest and most distributed network, characterized by several high in-degree hubs surrounded by many low-activity users, reflecting a broadcast-oriented discussion style. Alternative channels exhibit denser and more centralized networks, indicating stronger peer-to-peer debate and localized clusters of interaction. CNN displays the smallest and sparsest network. However, temporal analysis indicates a substantial drop in CNN's video publishing frequency during the sampling window (January 5–20, 2026). Consequently, the reduced network size may partially result from sampling bias due to API limitations and lower content availability, rather than solely reflecting audience passivity.

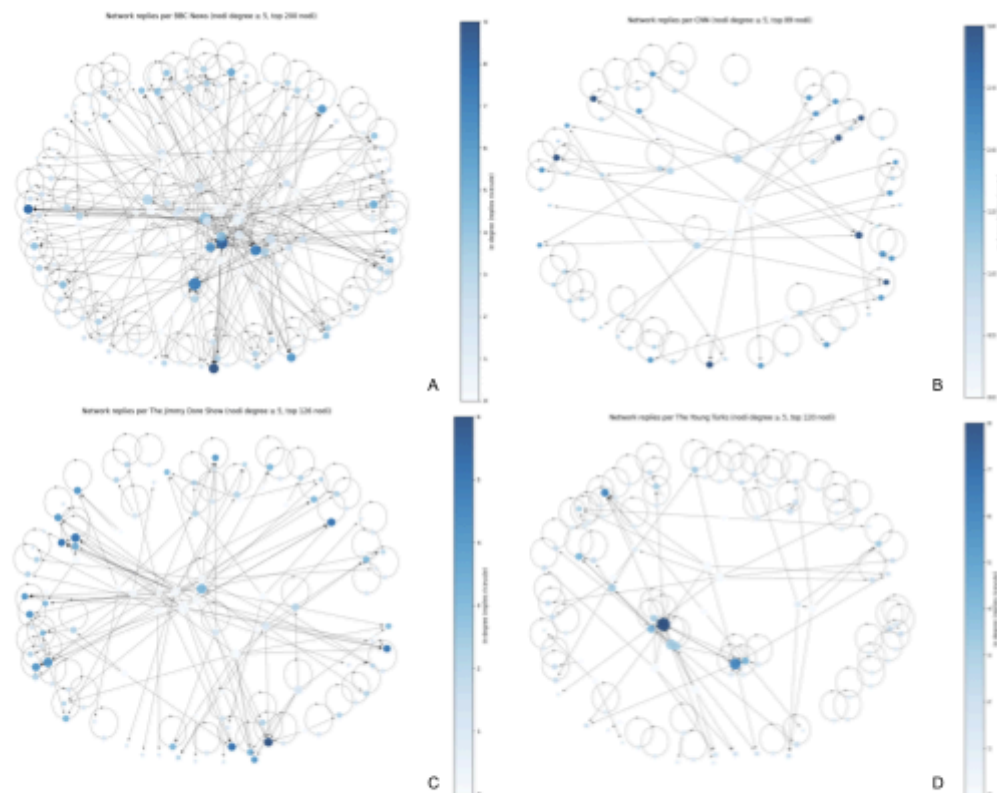


Figure 4: Reply network visualization by channel (nodes with degree ≥ 5). (a) BBC News; (b) CNN; (c) The Jimmy Dore Show; (d) The Young Turks; Node size: node degree; Colour Intensity: node In-degree

The findings challenge the assumption that surface-level clickbait alone drives engagement. Weak correlations between clickbait scores and comment volume suggest that users respond primarily to thematic framing, rather than formal sensationalism. Engagement is driven by topic salience, specifically polarizing political subjects, rather than the syntactic structure of the headline. The data suggest a category realignment with respect to the initial hypothesis, a three-tiered structure emerges:

- *institutional broadcaster* (BBC) with high stability, broad but centralized interaction, consistent with an informational mandate
- *commercial/partisan leaders* (CNN and TYT) with high volume but high volatility, relying on sensational or polarizing events to spike engagement
- *niche radical channel* (TJDS) characterized by intense but limited participation

Future research should expand the temporal sampling window to correct for potential activity fluctuations (as observed with CNN) and incorporate sentiment analysis of replies to further distinguish between supportive community building and polarized argumentation.

References

- Bakir, V., & McStay, A. (2017). Fake news and the economy of emotions. *Digital Journalism*, 6(2), 154–175. <https://doi.org/10.1080/21670811.2017.1345645>
- Blom, J. N., & Hansen, K. R. (2014). Click bait: Forward-reference as lure in online news headlines. *Journal of Pragmatics*, 76, 87–100. <https://doi.org/10.1016/j.pragma.2014.11.010>
- Boberg, S., Quandt, T., Schatto-Eckrodt, T., & Frischlich, L. (2020). Pandemic populism: Facebook pages of alternative news media and the corona Crisis -- A computational content analysis. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2004.02566>
- Chakraborty, A., Paranjape, B., Kakarla, S., & Ganguly, N. (2016). Stop Clickbait: Detecting and preventing clickbaits in online news media. *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 9–16. <https://doi.org/10.1109/asonam.2016.7752207>
- Mukherjee, P., Dutta, S., & De Bruyn, A. (2022). Did clickbait crack the code on virality? *Journal of the Academy of Marketing Science*, 50(3), 482–502. <https://doi.org/10.1007/s11747-021-00830-x>
- Pengnate, S. F., Chen, J., & Young, A. (2021). Effects of clickbait headlines on user responses: An Empirical investigation. *Journal of International Technology and Information Management*, 30(3), 1–18. <https://doi.org/10.58729/1941-6679.1440>

Appendix - Data Access

The datasets generated during the current study are available at:

https://drive.google.com/drive/folders/1AF38VDbmupFQCE-gFXKb6M0HwTPactNW?usp=drive_link

The repository includes raw data retrieved via the YouTube Data API and preprocessed files used for sentiment, emotion, and network analyses.